

1. How much work is done in emptying a conical tank full of water with radius 4 ft and height 10 ft by pumping the the water through a pipe extending 3 ft above the tank.
2. A rectangular gate built into a dam is 4 ft high and 8 ft wide. The top of the gate is 2 ft below the surface of the water. What is the force on the gate?
3. Set up an integral to find the volume of the solid generated when the region in the first quadrant bounded by $y = 4 - x^2$ and $y = 3$ is
 - (a) Rotated about the x -axis
 - (b) Rotated about the y -axis
 - (c) Rotated about the line $x = -3$
 - (d) Rotated about the line $y = -2$
 - (e) Rotated about the line $y = 5$
 - (f) Rotated about the line $x = 6$
4. Set up integrals to find the mass, moments, and center of mass of the region in the first quadrant bounded by $y = 4 - x^2$ and $y = 3$.
5. Find the area enclosed by three leaves of the four-leafed rose $r = 5 \cos(2\theta)$.

Evaluate the following integrals:

6. $\int \cot x \, dx$
7. $\int \sec x \, dx$
8. $\int \frac{1}{x} \, dx$
9. $\int \cos^2 x \sin x \, dx$
10. $\int e^{2x} \sinh(5x) \, dx$
11. $\int \frac{3}{x^2 + 4x} \, dx$
12. $\int \frac{\sin(\ln 3x)}{2x} \, dx$